



**McCausland College
of Arts and Sciences**
UNIVERSITY OF SOUTH CAROLINA

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Computational Analysis of Cell Fate Transitions in Alzheimer's Disease Mouse Model Using Single-Cell RNA Sequencing Data

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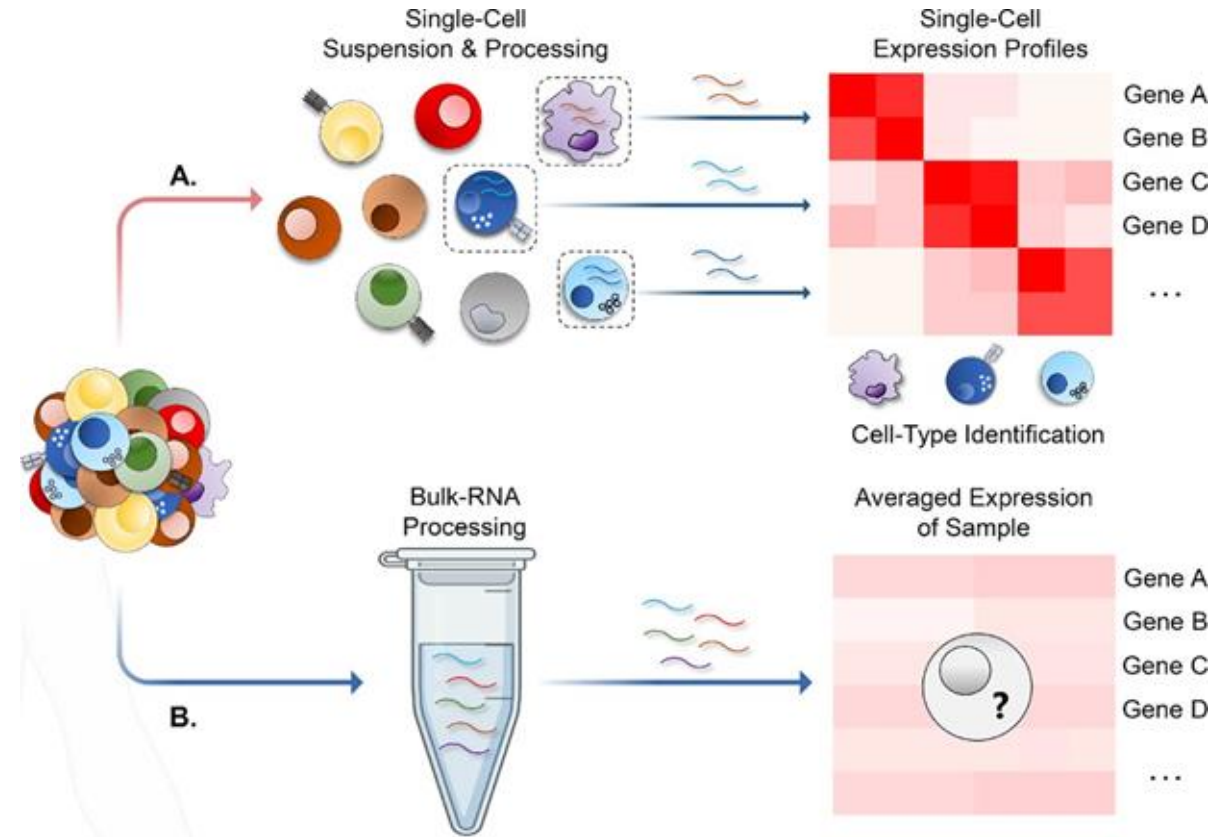
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Part I:
**Introduction to Methods and Cell
Trajectory Analysis**

Introduction to single-cell RNA sequencing data

- Single-cell RNA sequencing (scRNA-seq)
 - Profiles gene expression at the individual cell level
 - Reveals cellular heterogeneity and identifies distinct cell populations
 - Enables reconstruction of developmental and disease-related cellular states
- Bulk RNA sequencing
 - Average expression across all cells
 - Require deconvolution process for single-cell or cluster-level resolutions



Parse Biosciences, 2023

Research Objectives

1. Identify cell-cell communication pathways
1. Determine cell fate trajectory
1. Develop adjusted method for cell communication strength

Overview of data processing and analysis procedures

- **Seurat Analysis**

- Quality control, normalization, and feature selection
- Dimension reduction with PCA
- Clustering based on similar gene expression



- **CellChat Analysis**

- Uses annotated cell types from Seurat to infer ligand–receptor interactions
- Dimension reduction based on communication network similarity
- Identifies communication patterns and functional relationships between cell types



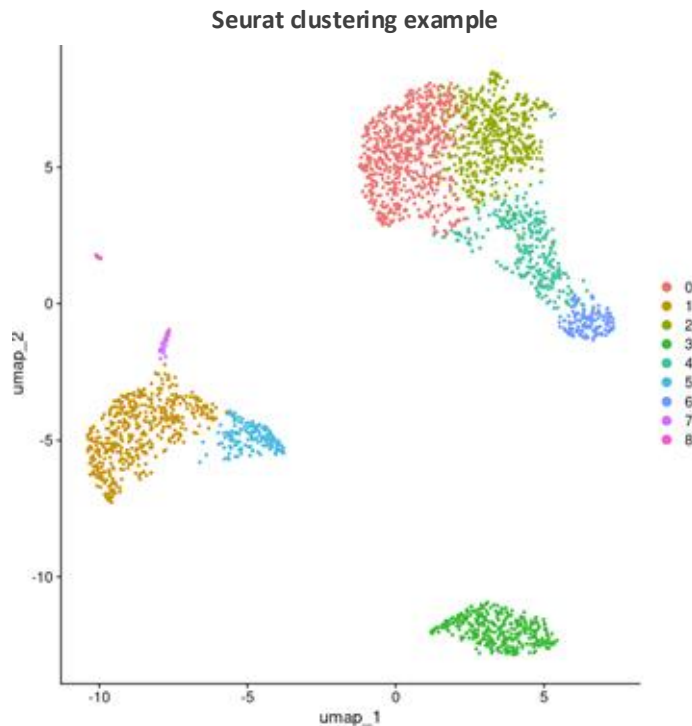
- **Monocle3 Analysis**

- Uses preprocessed/annotated cells to reconstruct developmental trajectories
- Trajectory inference based on transcriptomic similarity
- Pseudotime analysis: Assigns each cell a relative position along a trajectory to model dynamic biological processes

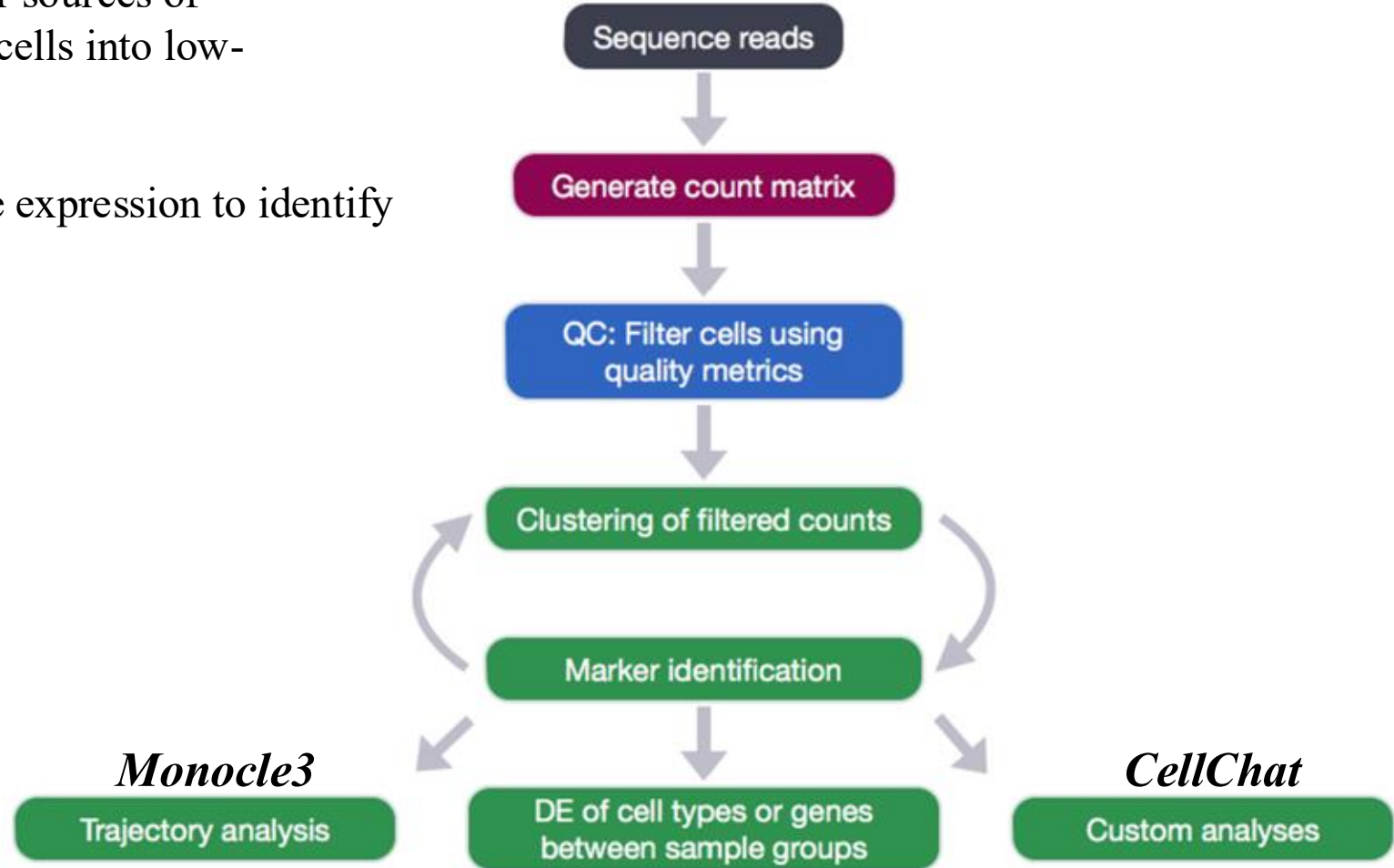


Processing & analyzing the scRNA-seq data using Seurat

- Dimension reduction: PCA captures major sources of transcriptomic variation; UMAP projects cells into low-dimensional space for visualization.
- Clustering: Groups cells with similar gene expression to identify distinct cell populations

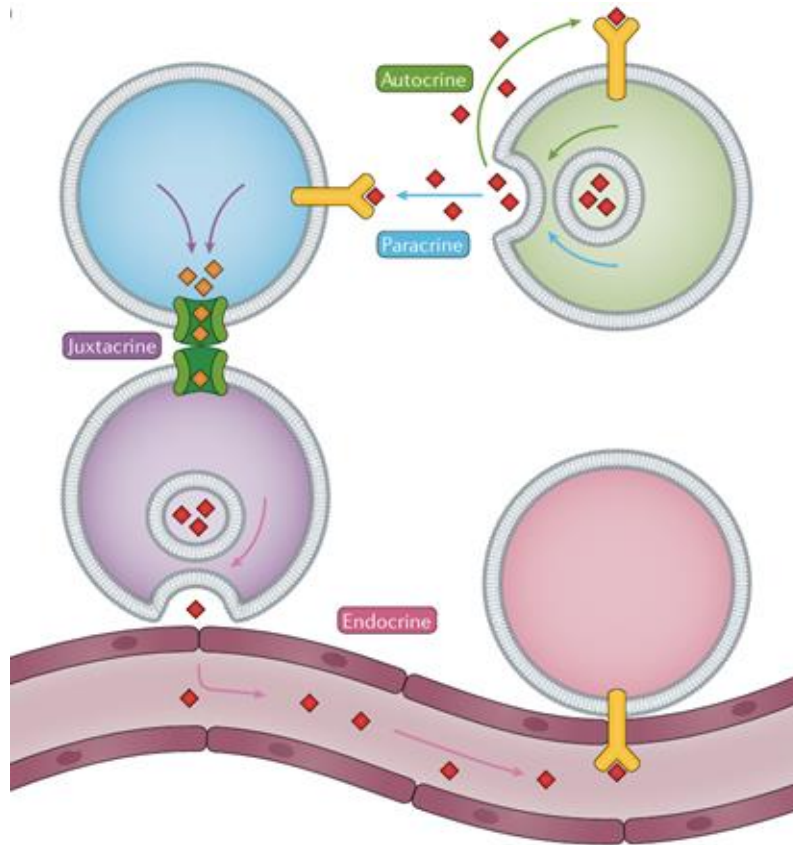


<https://hbctraining.github.io/>

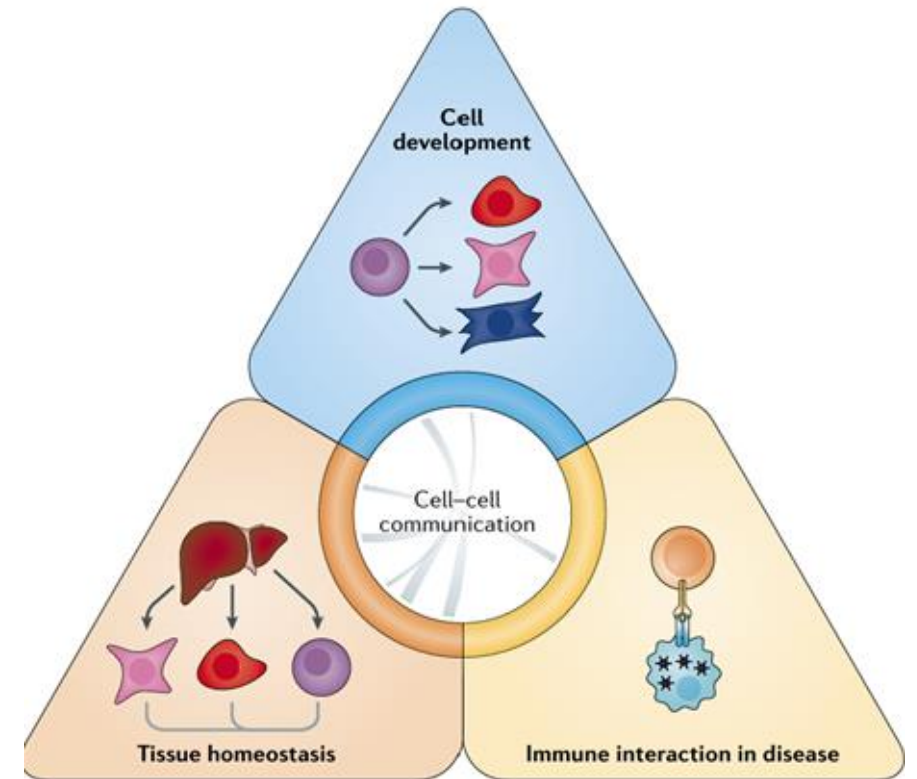


Cell-cell communication drives cell-fate transitions

Eric, et. al., Nature Reviews Genetics, 2021



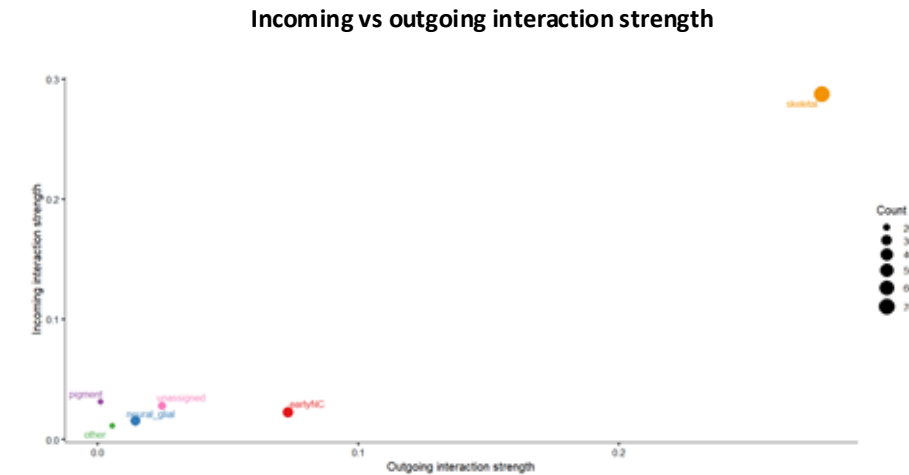
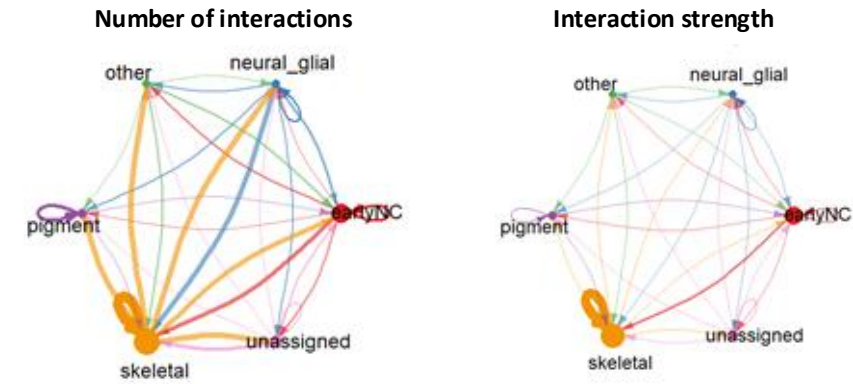
- Cell-Cell Communication considered in this study:
 - Cells communicate through ligand–receptor binding



- Cell Fate Transition:
 - Process by which cells change from one identity or functional state to another

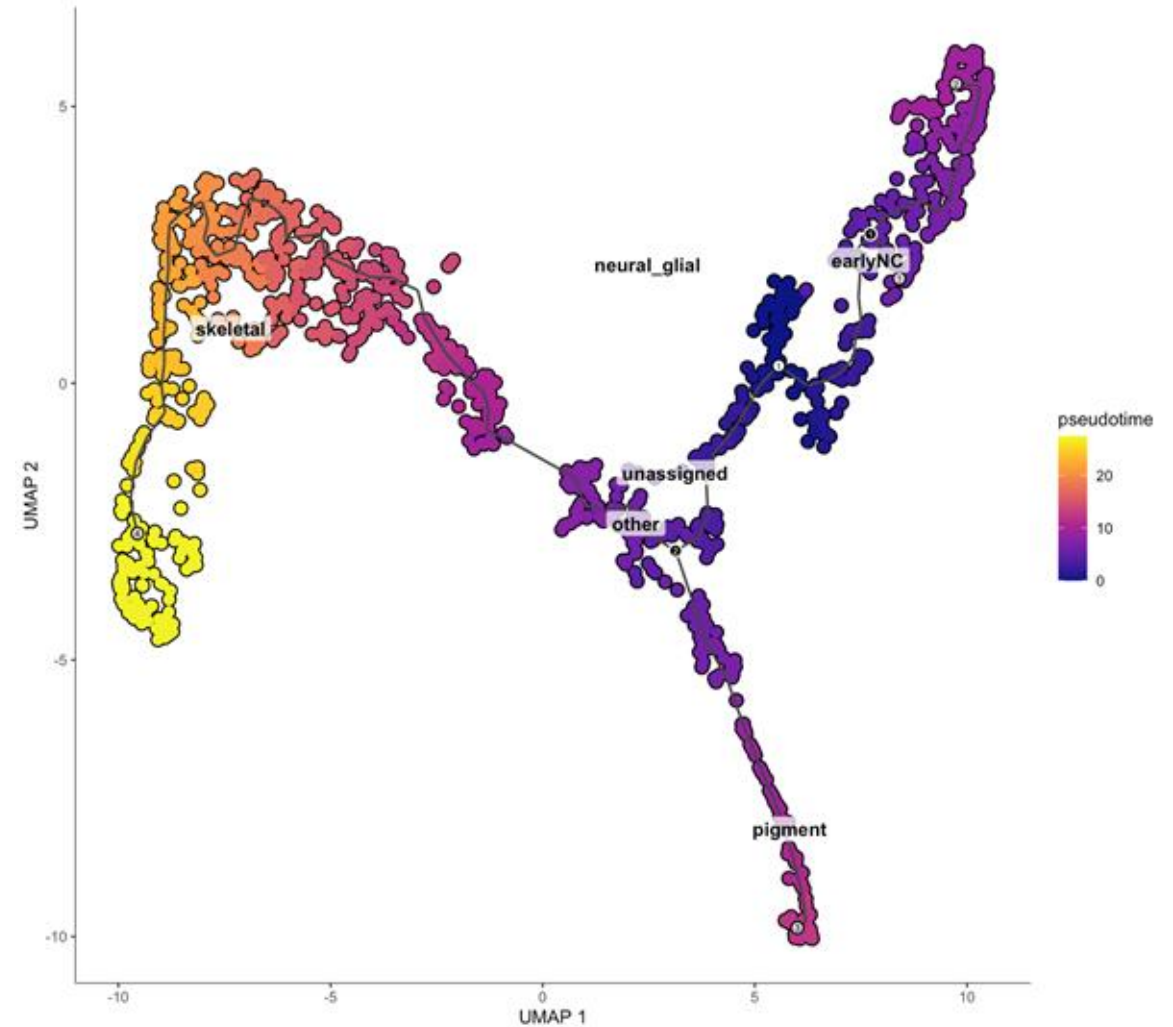
Overview of cell-cell communication inference based on ligand-receptor interactions

- **Building on Seurat**
 - Uses the annotated Seurat object (normalized expression matrix + cell type labels)
 - Treats identified cell clusters as sending and receiving cell populations
- **CellChat Workflow**
 - Identifies expressed ligands and receptors using a curated ligand–receptor database
 - Computes the probability and strength of communication between cell populations
 - Aggregates individual interactions into signaling pathways and communication networks
- **Outputs**
 - Cell–cell communication networks
 - Signaling pathway activity and interaction strength
 - Identification of major signal senders, receivers, mediators, and influencers



Pseudotime analysis revealing cell development trajectory

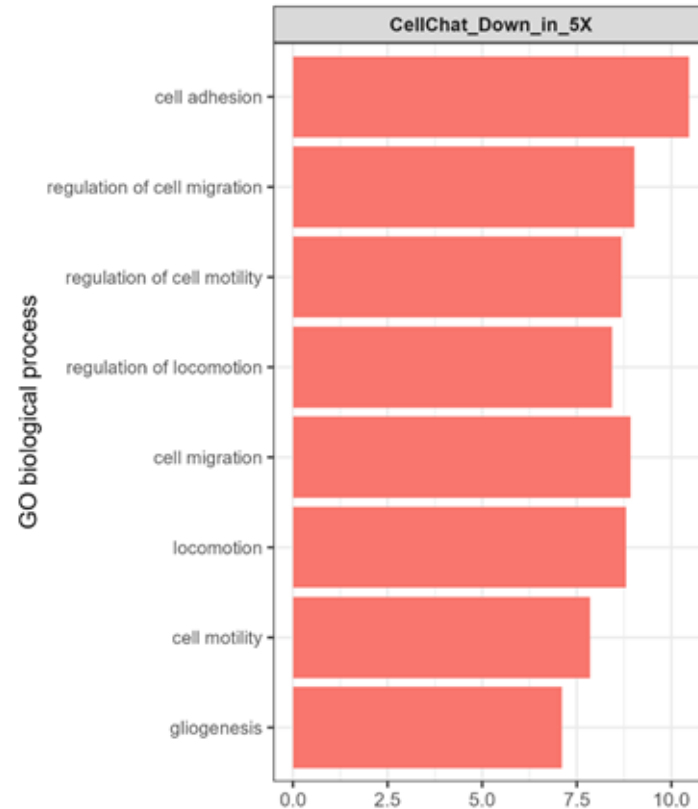
- Arranges single cells along an inferred path of change (**pseudotime**)
- Paths show movement of cells from one cell state to another
- Using the **Monocle3** package in R, we can analyze how specific cells may progress toward disease-associated states
 - Can perform trajectory-aware analysis on different cell stages (early, middle, late)



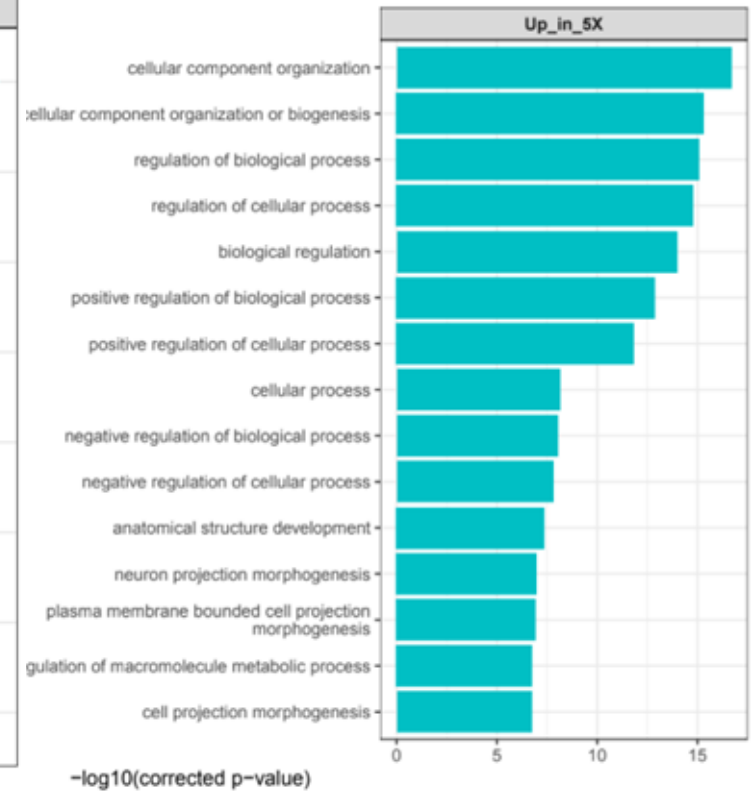
Identify top-associated biological processes via GO-term analysis

- Labels genes with **functional terms**
- Used to determine which biological programs are statistically overrepresented.
- Using the **gprofiler2** package in R, we can analyze how specific biological processes may change in disease-related states
 - Can see which genes are enriched
 - Used on differentially expressed genes (DEGs)

GO enrichment of CellChat-derived genes



GO enrichment of AD-related genes



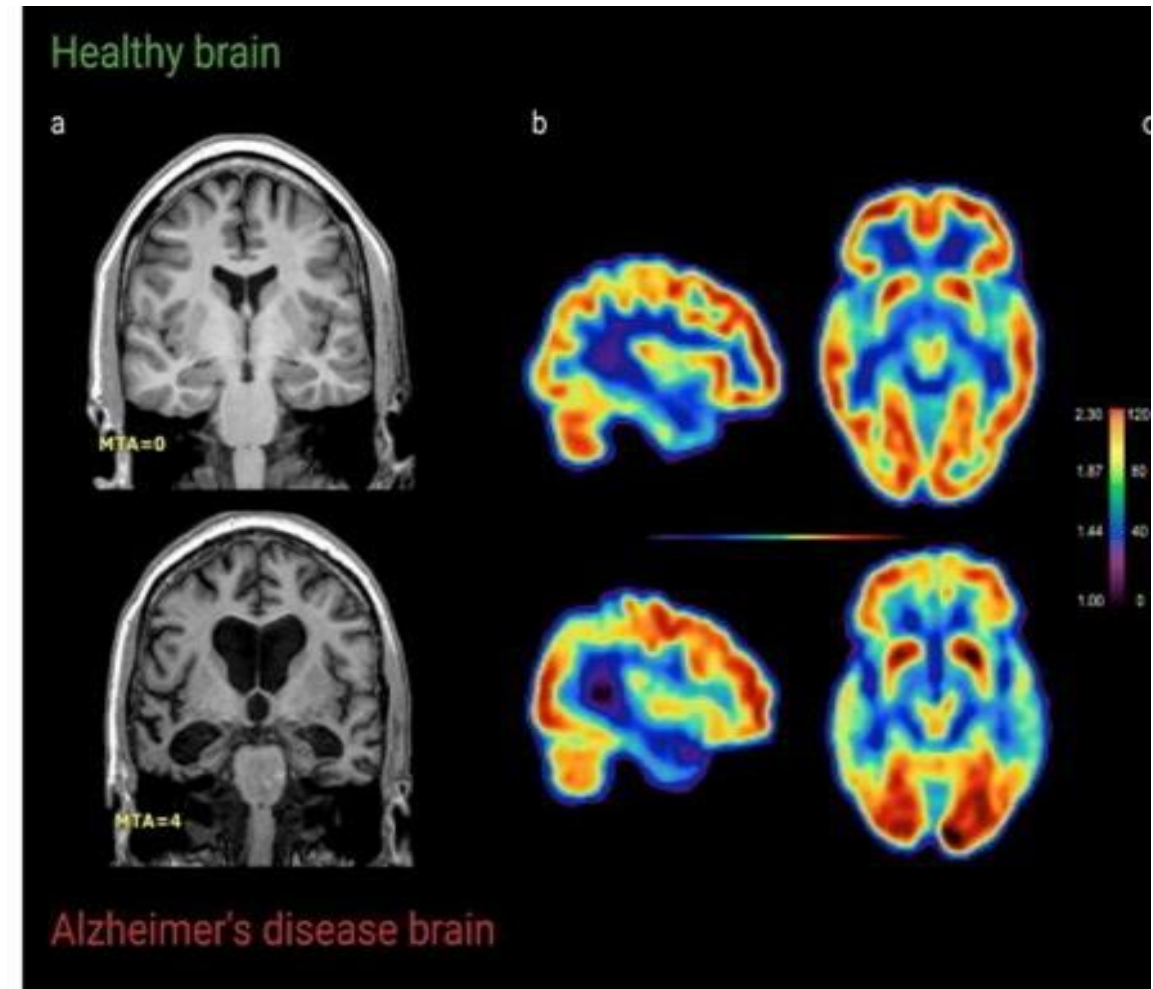
Single-cell data of mouse Alzheimer's Disease (5xFAD model)

Alzheimer's Disease (AD)

- Causes neuron damage and death, loss of brain tissue, and changes in cell gene expression
- Different cell types respond differently

5xFAD Data (*Zhou, et al., Nat Med, 2020.*)

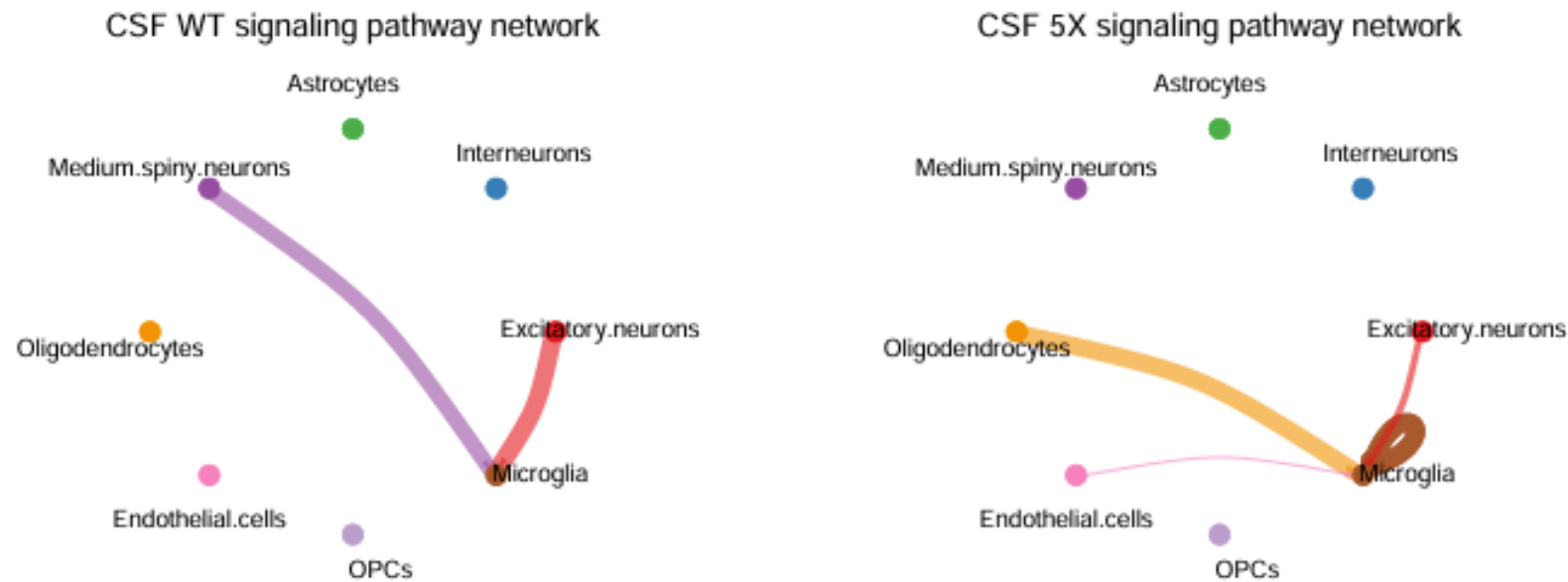
- Mouse model genetically engineered to express five (5) mutations associated with AD
- Studied in paper using scRNA-seq to examine brain cell populations in AD and the role of TREM2
- Inferring AD-related communications through the following analysis:
 - Cell-cell communication (**CellChat**)
 - Gene ontology enrichment analysis (**gprofiler2**)
 - Trajectory-aware pseudotime (**Monocle3**)



Ahmad et al., Acta Biomed, 2023.

Part II:
5xFAD Analysis: Cell-Cell
Communication and Trajectory Results

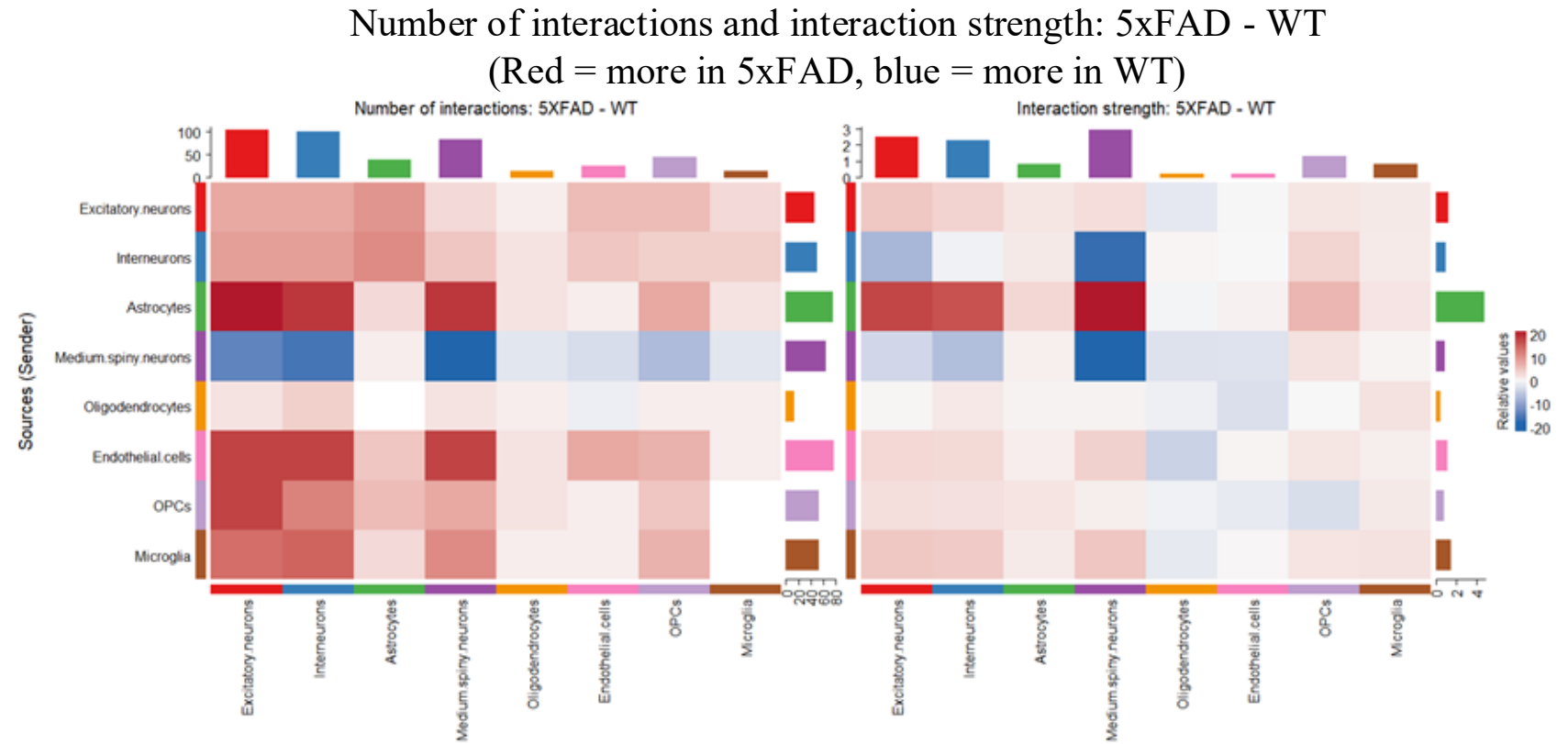
The significant change of Colony Stimulating Factor (CSF) signaling



- Colony Stimulating Factor 1 (CSF1) and its receptor (CSF1R) govern microglial survival, proliferation, and activation states (*Adrian, et al, Brian, 2016*).
- CSF1 released by distressed oligodendrocytes or oligodendrocyte precursor cells (OPCs) hyper-activates microglia, accelerating demyelination and exacerbating cognitive decline (*Kinga, et al, J. Neuroimmune Pharmacol., 2025*).

Glial and vascular-associated communication increases in 5xFAD

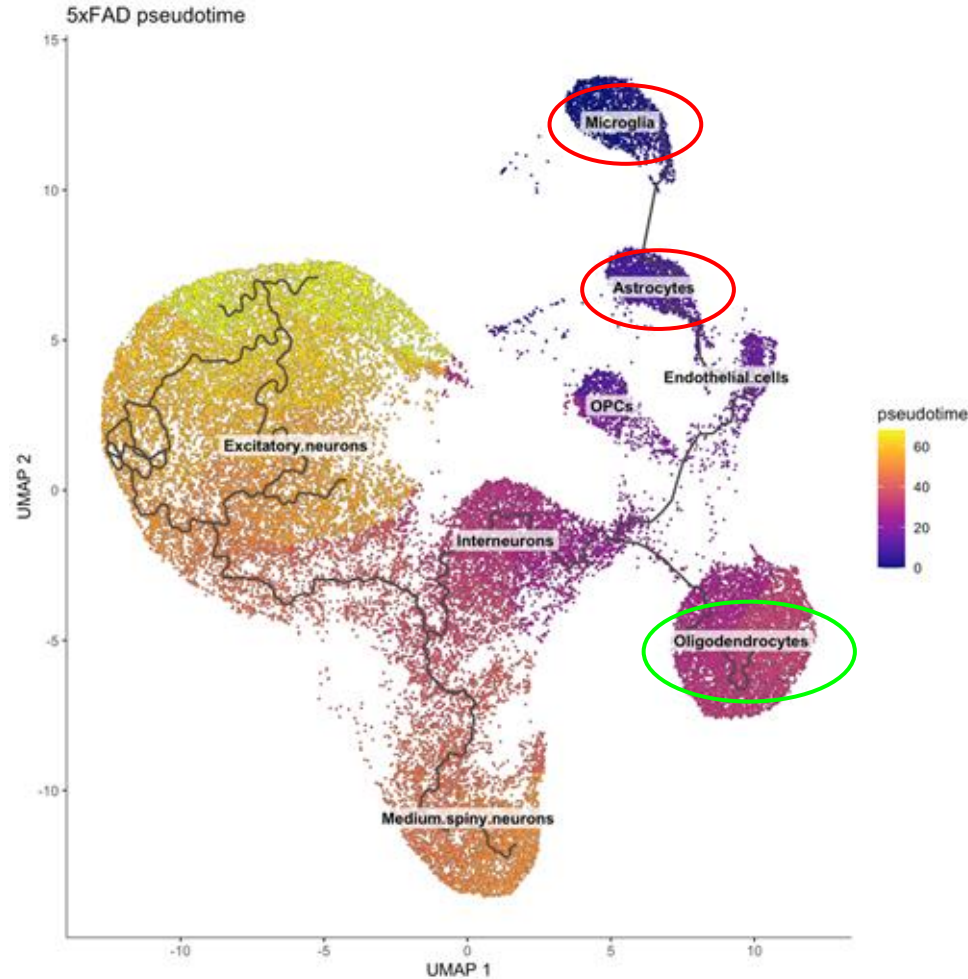
- Glial and vascular-associated cells (astrocytes, endothelial cells, OPCs, and microglia) show the largest increases in outgoing communication.
- Neuronal populations show mixed patterns, suggesting disease selectively rewires glial-neuronal signaling rather than uniformly upregulating everything.



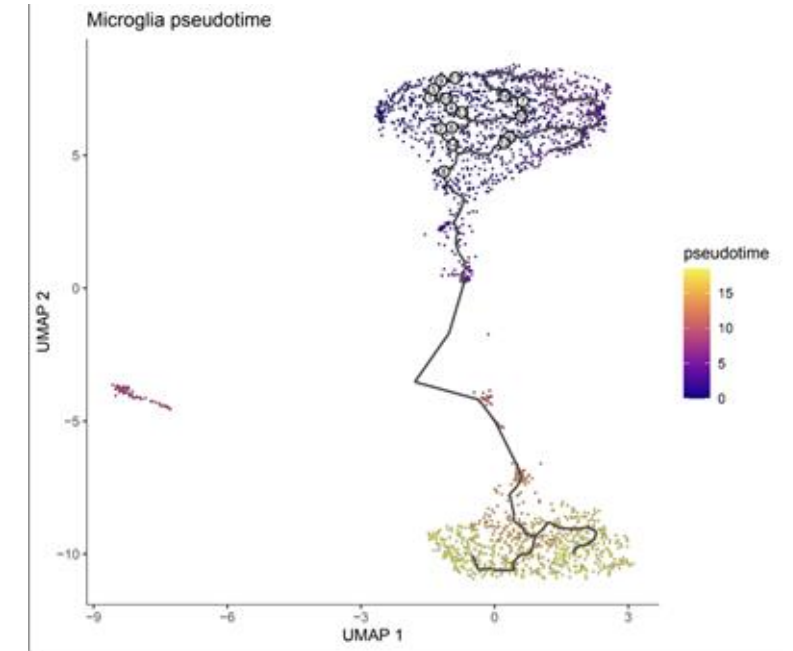
- What's going on in the brain: Astrocytes and microglia activate, releasing inflammatory markers; endothelial cells upregulate angiogenesis to compensate for vascular dysfunction; and Oligodendrocyte Progenitor Cells (OPCs) proliferate in a failed attempt to repair damaged myelin (*Peibin, et al, Transl Neurodegener, 2023*).

Cell development trajectory inferred by pseudotime analysis

- Glial cell types – microglia, astrocytes, and oligodendrocytes – cluster separately from neuronal populations, with microglia at the root and the trajectory flowing outward toward neurons.
- 5xFAD cells are enriched toward the terminal ends of each branch, suggesting disease pushes cells further along their existing trajectory.

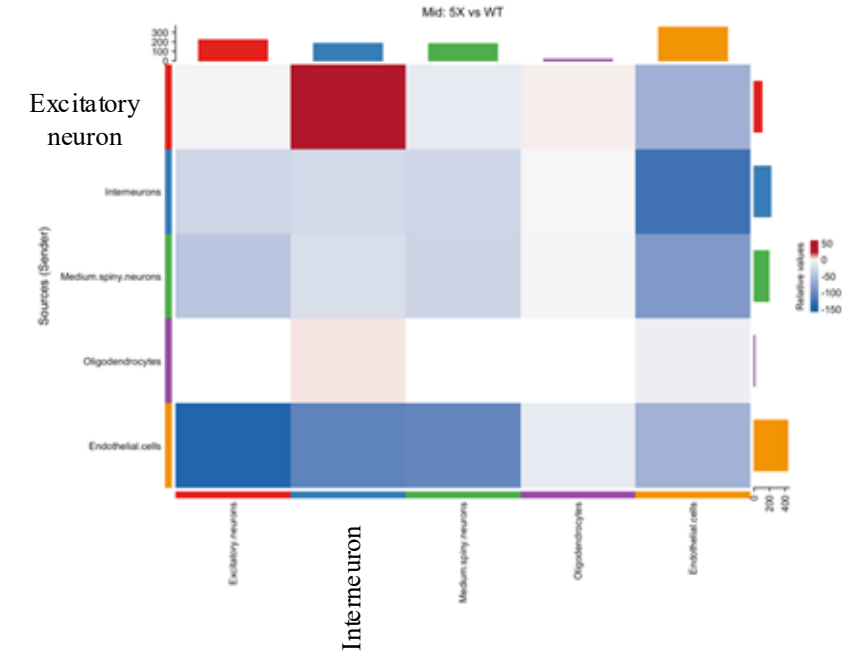
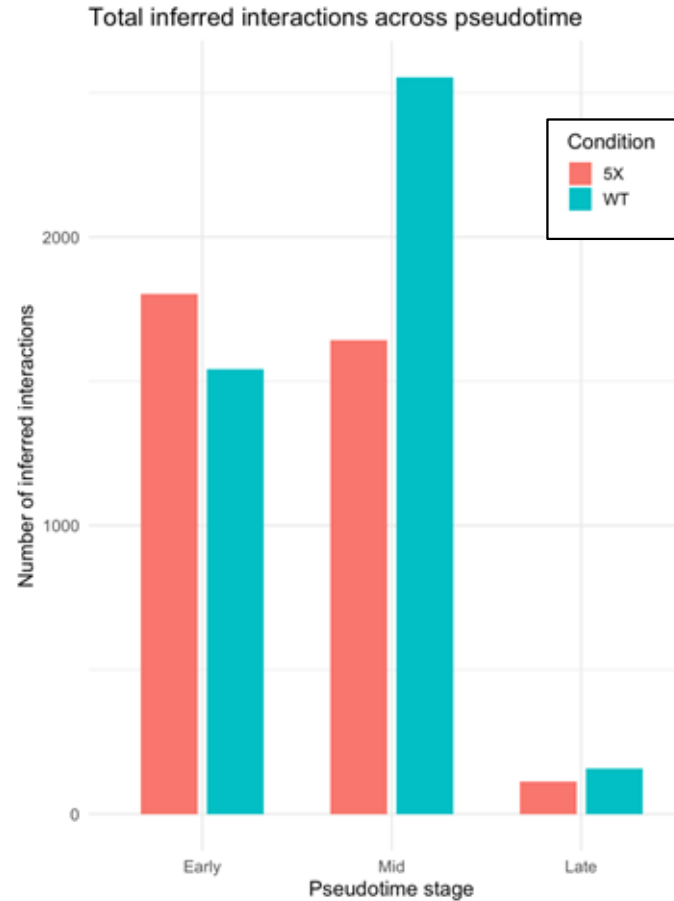


- Zooming into microglia reveals two subpopulations connected by a clear trajectory, likely representing the transition from homeostatic microglia to disease-associated microglia (DAM).



Trajectory-aware cell-cell communication analysis -- a new analyzing pipeline from existing methods

- Split cells into early, mid, and late pseudotime bins using natural breaks in the distribution, then ran CellChat separately on each bin and condition.
- 5xFAD starts with more total interactions than WT at the early stage, but WT overtakes it by mid stage before both drop sharply at late.



- Mid-stage differential heatmap shows WT pulling ahead broadly, driven largely by endothelial cell communication, while 5xFAD retains stronger excitatory neuron to interneuron signaling.

Part III:

Improved method to quantify the cross- condition change

Limitations of Current Methods

THE PROBLEM

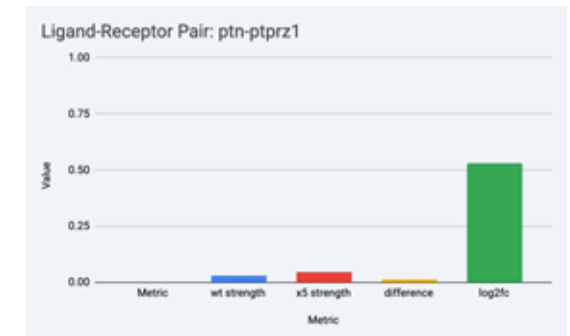
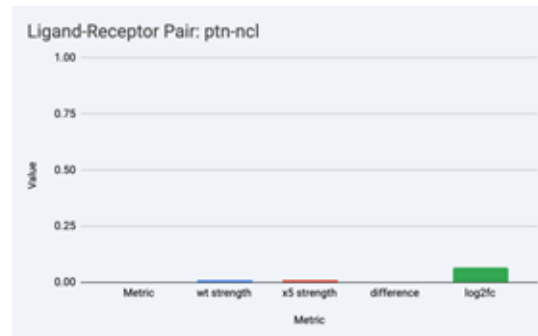
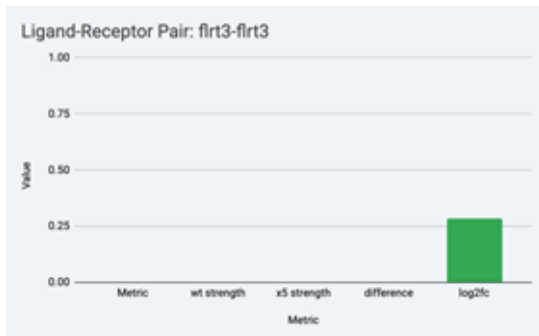
Current methods fail to quantify conditions precisely:

- They do not directly quantify changes between biological conditions.
- They lack consideration for statistical robustness, treating all communication strengths equally.
- Small, weak communication strengths are presented with the same magnitude as larger, highly robust ones.

THE GOAL

Develop an improved, strength-aware shrinkage method:

- Design a mathematical method to shrink Log2FC based systematically on underlying communication strength.
- Enable high-confidence identification of true disease-critical genes by shrinking weak or uncommon ligand-receptor pairs.



Communication Strengths Method Comparison

1. Adjusted Log2FC

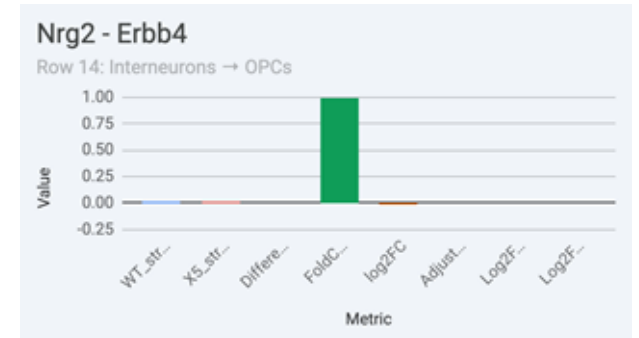
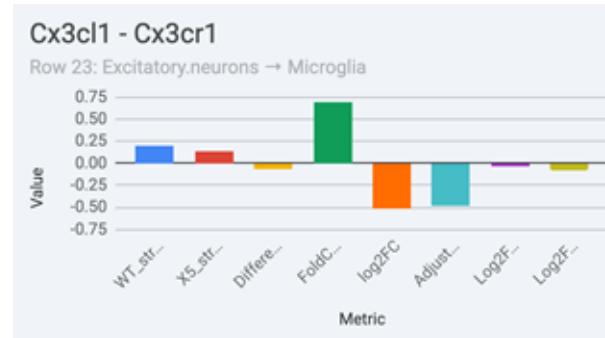
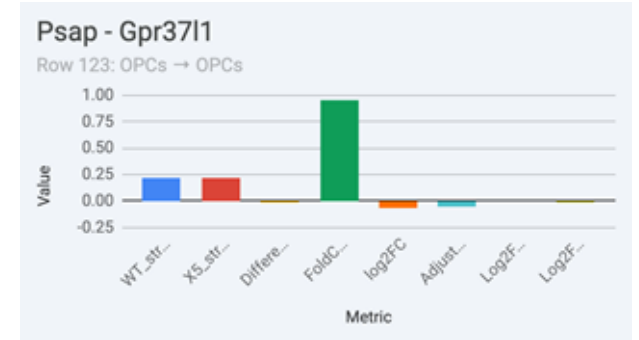
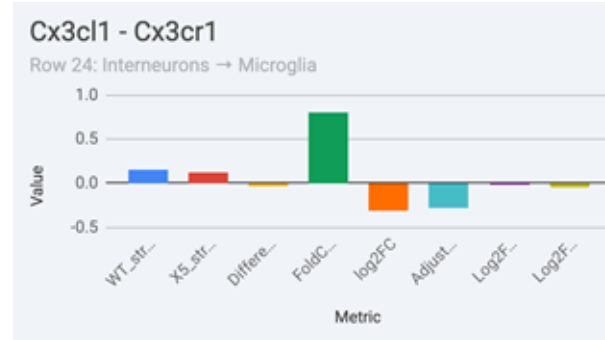
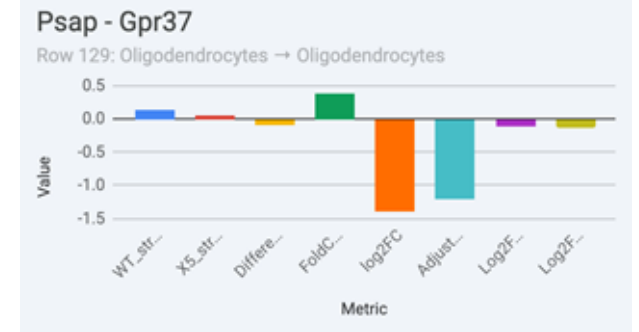
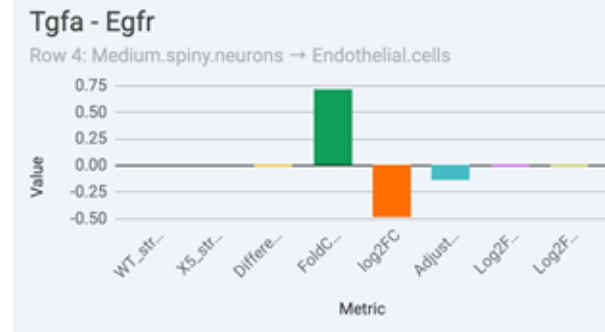
$$\log_2 \left(\frac{X5_strength + CommunicationFloor}{WT_strength + CommunicationFloor} \right)$$

2. Difference * Log2FC

$$(|Wtstrength - X5strength|) * \log_2 FC$$

3. Mean * Log2FC

$$\log_2 FC * \left(\frac{(WTstrength + X5strength)}{2} \right)$$

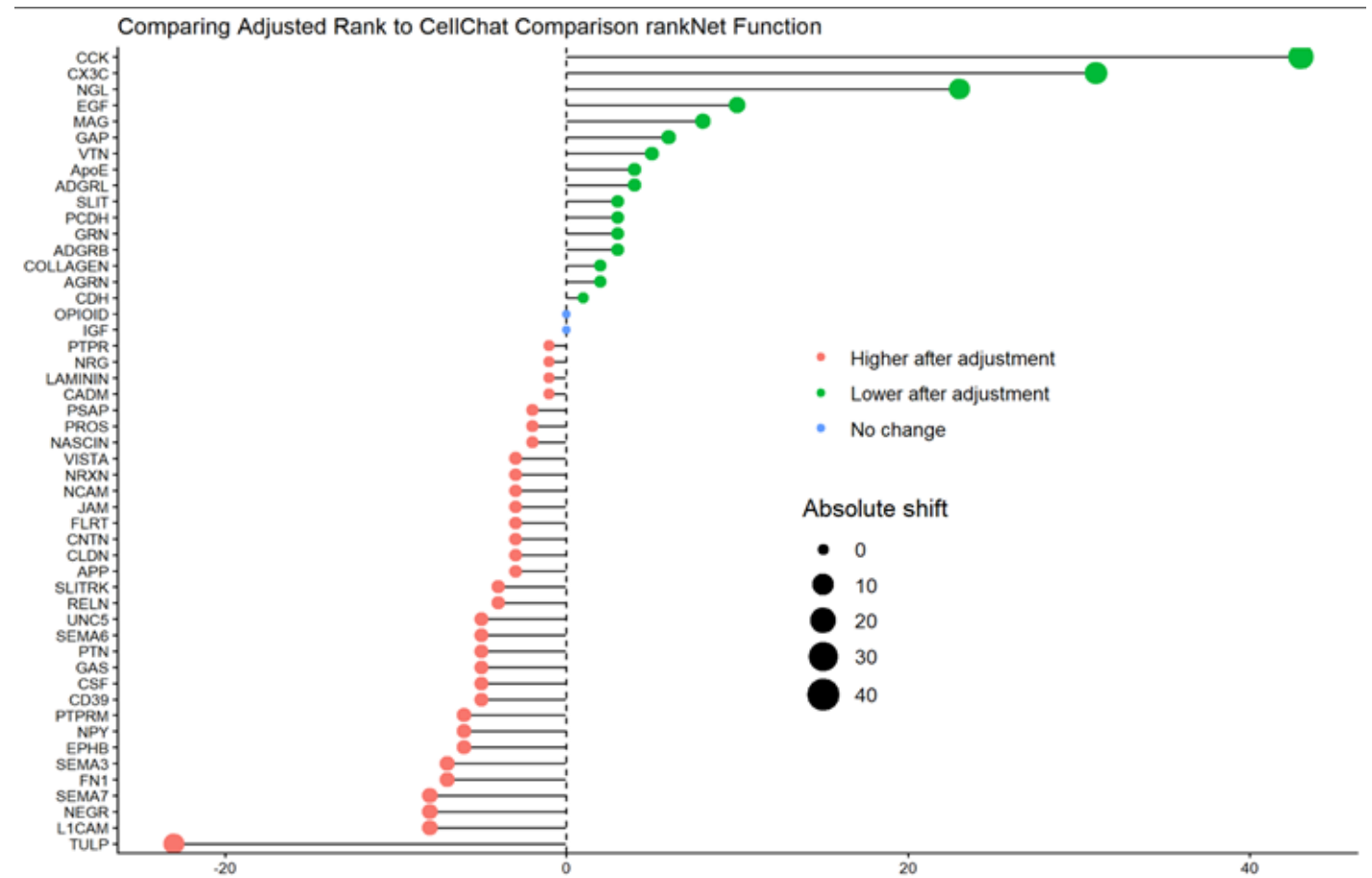


Adjusted Log2FC to identify critical disease-driver ligand-receptor pairs

Adjusted Log2FC =

$$\log_2 \left(\frac{X5_strength + CommunicationFloor}{WT_strength + CommunicationFloor} \right)$$

- Communication Floor is median communication strength of non-significant ligand-receptor interactions
- TULP genes increased due to high baseline expression



WT_strength	X5_strength	Difference	FoldChange	log2FC	Adjusted_log2FC
0.00516888826	0.003689220666	-0.00147966759	0.7137358138	-0.486537929	-0.1306810321
0.2021712102	0.141560594	-0.06061061625	0.700201546	-0.5141578476	-0.4801762932
0.1520069551	0.1225846049	-0.02942235027	0.8064407629	-0.3103595308	-0.2854025868

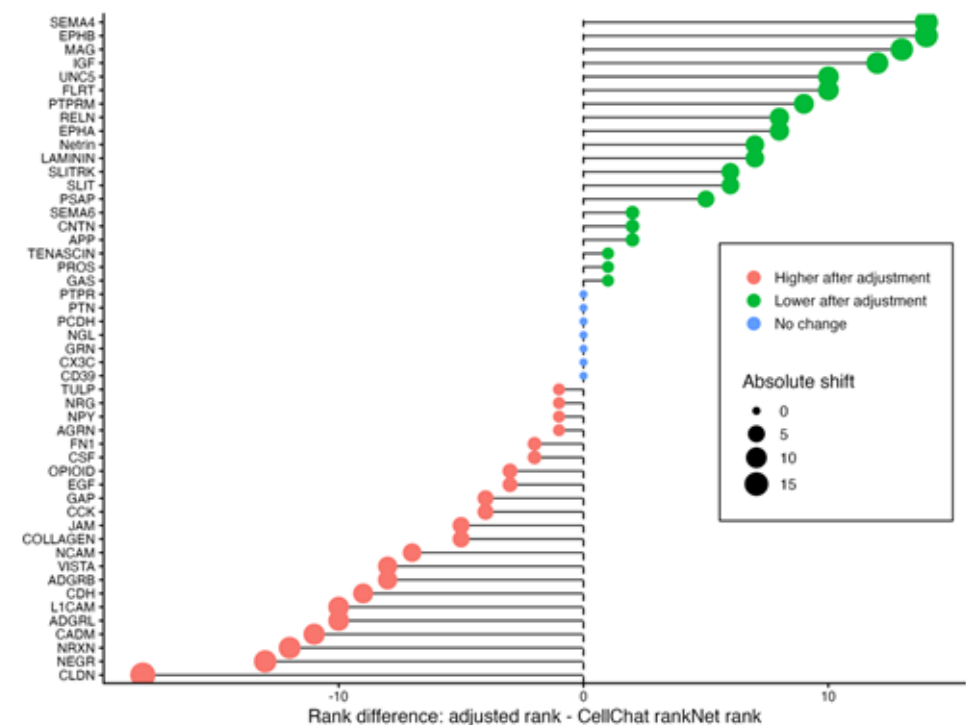
Incorporating Bootstrap Results into Adjusted Log2FC

Tested various ways of integrating bootstrap results into adjusted Log2FC

1. SNR Score Method

$$\frac{\text{AdjustedLog2FC}}{SD + 0.1}$$

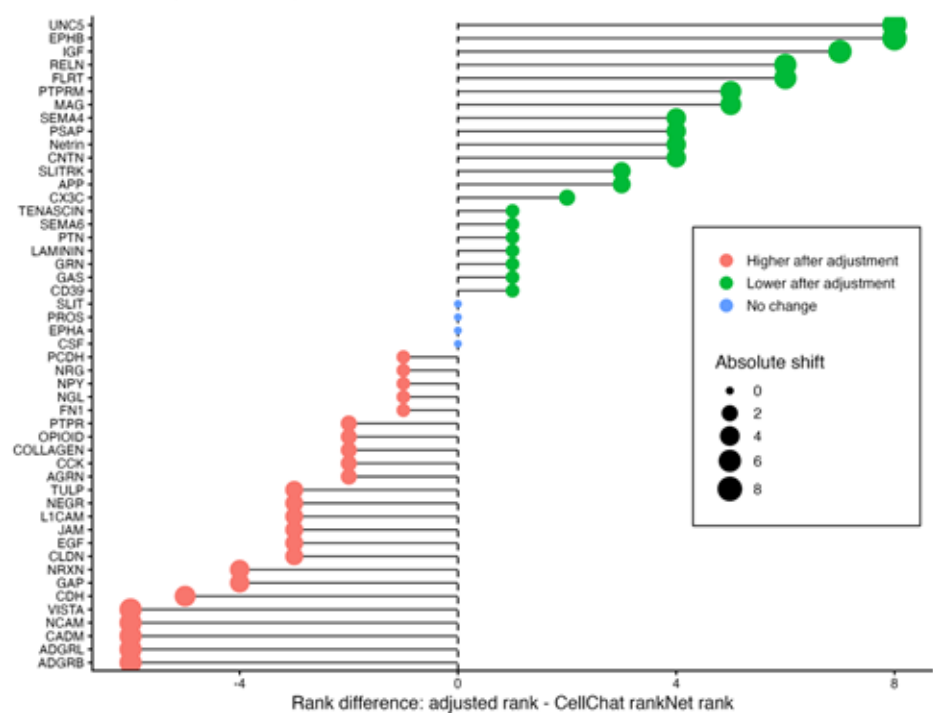
Comparing SNR Score to CellChat Comparison rankNet Function



2. Bootstrap Log2FC Method

$$\text{AdjustedLog2FC} * \frac{\text{Penalty}}{SD + \text{Penalty}}$$

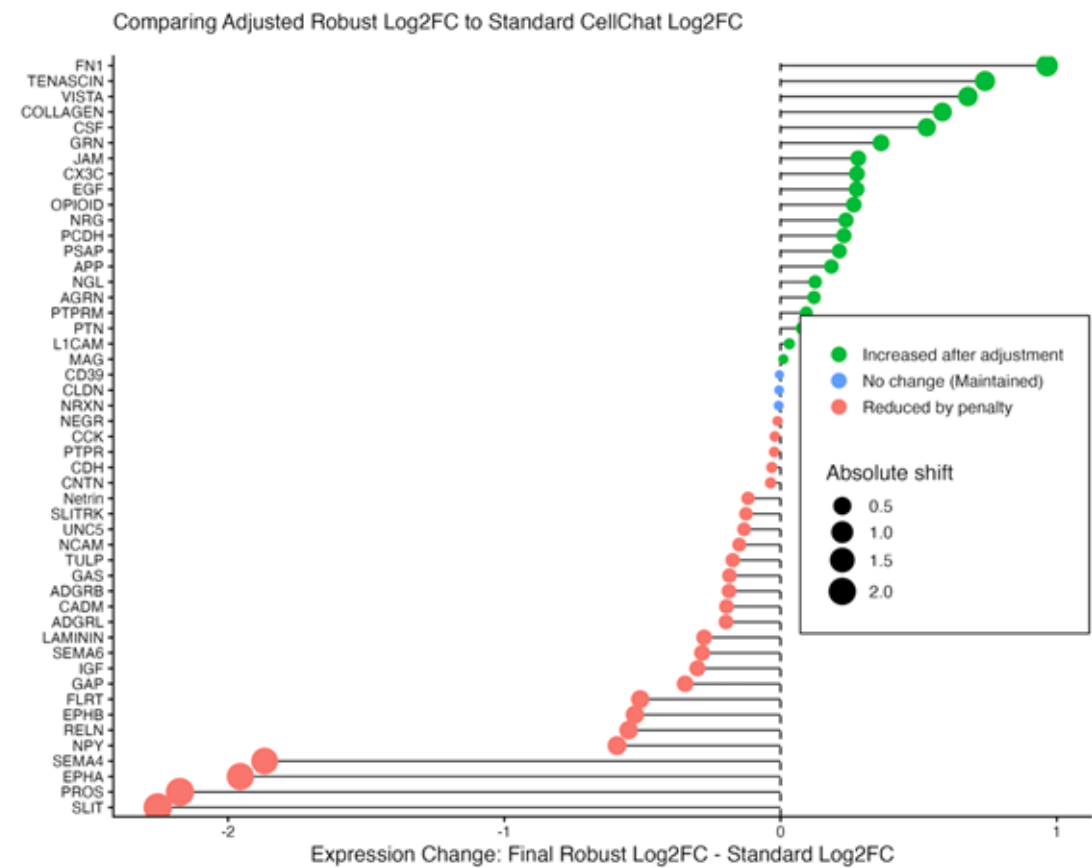
Comparing Bootstrap Log2FC to CellChat Comparison rankNet Function



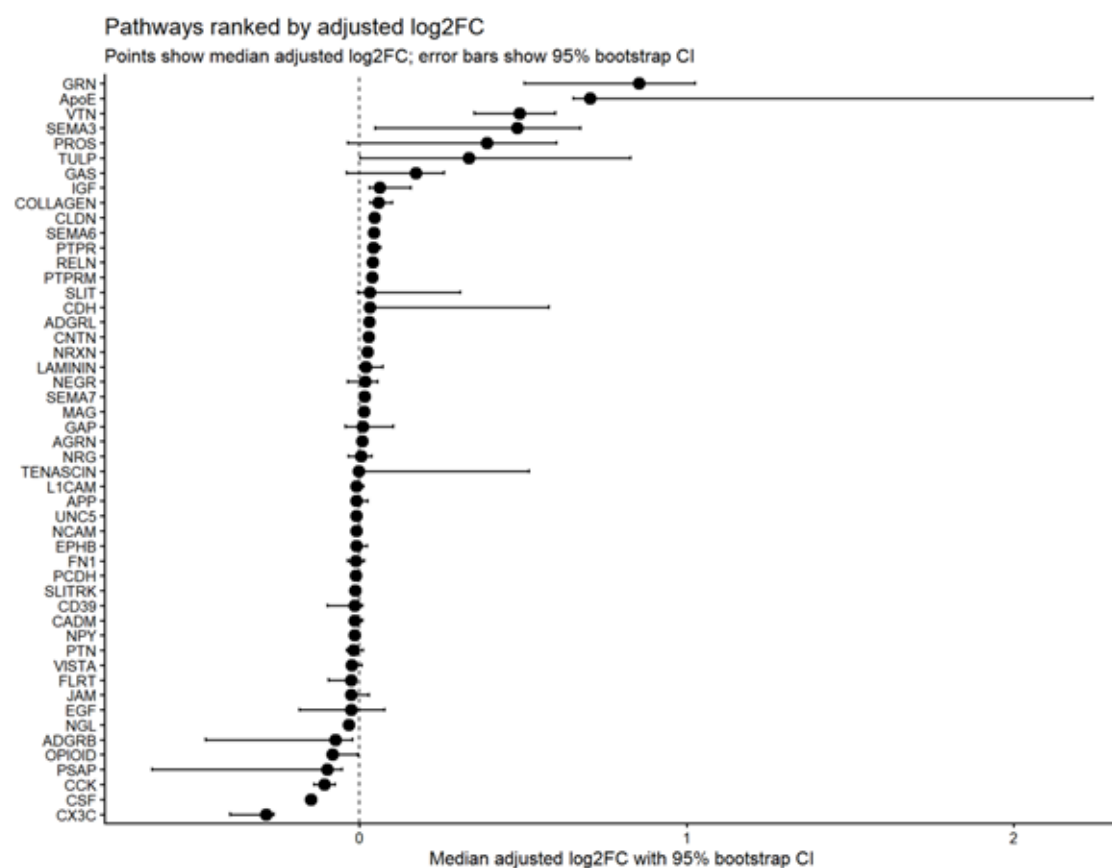
Final Methods Evaluated

New methods incorporate communication strength into Log2FC, successfully emphasizing critical gene groups

1. Expression Change (Standard vs. Robust)



2. Pathway Rankings & Confidence Intervals



Summary

- Successfully combined Seurat clustering, CellChat network analysis, gprofiler2 GO enrichment analysis, and Monocle3 trajectory tracking to study scRNA-seq transitions in the 5xFAD Alzheimer's Disease mouse model
- Identified that 5xFAD cells are pushed further toward the terminal ends of lineage trajectories. Zooming into microglia revealed a distinct transitional arm capturing the progression from a homeostatic state to a Disease-Associated Microglia
- Breaking trajectories down into early, mid, and late stages demonstrated that microglia-specific signaling steadily increases across disease pseudotime, consistently overtaking wild-type baselines
- Developed a strength-aware mathematical shrinkage method utilizing a data-derived CommunicationFloor. This correction successfully penalizes low-intensity, noisy interactions while preserving high-confidence biological drivers

Future Research Plan

- Investigate the efficacy of a Variational Autoencoder (VAE) for running scRNA pseudotime analysis and incorporate spatial data
 - Stochastic decoder assumes a Zero-Inflated Negative Binomial distribution for raw count data
 - Posterior distribution is approximated by a Gaussian
 - Compare with PCA and ICA dimension reduction
- Verifying new communication gene results with Open Targets prior data.
 - Explore incorporating existing gene-disease association data as statistical priors
- Continue to look at various methods for log2FC communication strength

Acknowledgements

- 2026 REU Program: Mathematical Foundation of Data Science for hosting this program and providing the environment to make our research possible.
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- Jasdeep Singh for providing foundational lectures and offering strategic advice on navigating complex blocks of our research.

Thank you!

Research Team

